

STATE BANK OF INDIA

Risk Management Department

Enterprise Level Unified
Risk Management Dashboard

COMPREHENSIVE TECHNICAL & ANALYTICAL REFERENCE GUIDE

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1. Executive Summary

Purpose of this Document. This reference guide provides a complete technical and analytical walkthrough of the SBI Enterprise Risk Management Dashboard. It is written to be fully self-contained: a reader with no prior risk management background should be able to understand every metric, every chart, and every alert in the dashboard after reading this document.

What the Dashboard Does. The dashboard consolidates data from more than a dozen internal source systems — from CBS (Finacle) to the ALM System, SIEM, and Climate Risk Tool — into a single, real-time view. It covers all major risk categories required under the Basel III / RBI regulatory framework, organised into three Basel Pillars.

1.1 The Three Basel Pillars

The Basel III framework (adopted by RBI in India) divides banking regulation into three pillars, each serving a distinct purpose:

Pillar	Name	What it Covers
Pillar 1	Minimum Capital Requirements	Credit Risk, Market Risk, Operational Risk — the "hard" capital the bank must hold by law.
Pillar 2	Supervisory Review Process	Liquidity, IRRBB, Climate, Cyber, Fraud, Concentration, Reputation — risks that fall outside Pillar 1 but are monitored and stress-tested.
Pillar 3	Market Discipline	Public disclosures on capital, leverage, and risk — builds transparency with investors and regulators.

The dashboard is built around **18 worksheets** covering all three pillars, supported by a central Monthly_Data source sheet that feeds every calculation automatically.

1.2 RAG Status — How Alerts Work

Every Key Risk Indicator (KRI) is assigned a traffic-light status based on its current value versus pre-approved thresholds:

Colour	Label	Meaning
 GREEN	SAFE / OK	KRI is within approved risk appetite. No action required.
 AMBER	WATCH	KRI is in a warning zone. Risk officers must monitor closely and prepare mitigation plans.
 RED	BREACH	KRI has breached the risk limit. Immediate escalation and corrective action is mandatory.

1.3 Dashboard Health Snapshot (April 2026)

As of the current reporting period, the overall portfolio risk profile is:

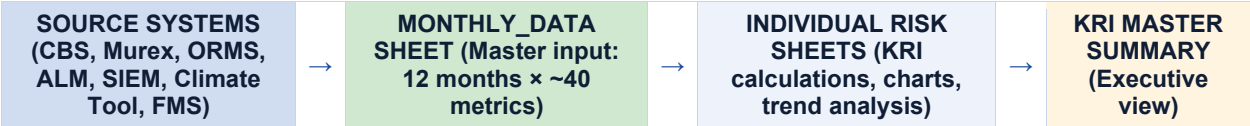
Metric	Current Status
Total KRIs Monitored	69 KRIs across 14 risk categories
KRIs in GREEN (Safe)	~48 KRIs (70%)
KRIs in AMBER (Watch)	~16 KRIs (23%)
KRIs in RED (Breach)	5 KRIs (7%) — Requires Immediate Action
Red KRI List	RCSA Score, HQLA Buffer, Single Borrower Limit, Fraud Recovery Rate, [Additional — see detail]

2. Data Architecture — How the Dashboard is Built

The dashboard follows a **single-source-of-truth architecture**. All raw numbers are entered only once in the **Monthly_Data** sheet. Every KRI in every other sheet is computed automatically via formulas that reference this single sheet. This prevents data inconsistencies and makes auditing straightforward.

2.1 Data Flow Diagram (Conceptual)

The data flows in the following sequence:



2.2 Monthly_Data Sheet — Complete Column Dictionary

This is the foundational sheet. Understanding each column is essential before reading any KRI chart. The table below defines every field.

2.2.1 Credit Risk Fields

Column Name	Unit	Definition & Formula Logic
Gross NPA Amount	₹ Crore	Sum of all loans where the borrower has defaulted for 90+ days (Days Past Due ≥ 90). Formula: SUMIF(NPA_Category ≠ "Standard", Outstanding_Balance)
Gross Advances	₹ Crore	Total outstanding loan book — the entire sum lent out by the bank across all segments. Formula: SUM(Outstanding_Balance for all loans)
Net NPA Amount	₹ Crore	Gross NPA minus provisions already set aside. Provisions are the money the bank has already reserved to cover expected losses. Formula: Gross_NPA – Provisions_held_against_NPA
SMA-0 Balance	₹ Crore	Special Mention Account stage 0: loans where the borrower is 1–30 days overdue. This is the earliest warning signal. Formula: SUMIF(SMA_Flag="SMA-0", Outstanding)
SMA-1 Balance	₹ Crore	SMA stage 1: loans 31–60 days overdue. Escalating risk. Formula: SUMIF(SMA_Flag="SMA-1", Outstanding)
SMA-2 Balance	₹ Crore	SMA stage 2: loans 61–90 days overdue. One more missed payment and the loan becomes NPA. Formula: SUMIF(SMA_Flag="SMA-2", Outstanding)
Total Provisions	₹ Crore	The total money set aside by the bank to absorb future credit losses across all loan categories. Computed from ECL (Expected Credit Loss) model under Ind AS 109.
ECL Stage 3 EAD	₹ Crore	Exposure At Default for Stage 3 (credit-impaired) loans under the Ind AS 109 ECL model. Equals the balance outstanding at the point of expected default.
ECL Stage 3 Provisions	₹ Crore	Provisions specifically held against Stage 3 exposures. ECL = EAD × PD × LGD (see formulas section).

2.2.2 Market Risk Fields

Column Name	Unit	Definition & Formula Logic
Trading Book VaR 1D	₹ Crore	Value at Risk for the trading portfolio over a 1-day holding period at the 99th percentile confidence level. Means: "On 99 out of 100 trading days, the bank will NOT lose more than this amount."

VaR Limit	₹ Crore	The maximum VaR amount approved by the Board. Current limit: ₹1,000 Cr. If VaR exceeds this, positions must be reduced immediately.
SVaR Total	₹ Crore	Stressed Value at Risk — VaR calculated using historical data from a severely stressed period (typically the 2008 financial crisis). Always higher than regular VaR. Used for regulatory capital calculation.
IRC Total	₹ Crore	Incremental Risk Charge: captures migration and default risk of credit-sensitive positions (bonds, CDSs) in the trading book over a 1-year horizon at 99.9% confidence.
FX Open Position	₹ Crore	Net unhedged foreign currency exposure. A positive number means the bank has more foreign currency assets than liabilities (long position). Regulated by RBI/FEMA.
DV01 Total	₹ Lakhs	Dollar Value of 01 — the change in the portfolio value for a 1 basis point (0.01%) increase in interest rates. A DV01 of ₹5,200 Lakhs means a 1bp rate rise costs ₹52 Crore.

2.2.3 Capital Adequacy Fields

Column Name	Unit	Definition & Formula Logic
CET1 Capital	₹ Crore	Common Equity Tier 1 — the highest quality capital: equity shares + retained earnings. This is the "first line of defence" against losses.
Tier 1 Capital	₹ Crore	CET1 + Additional Tier 1 (AT1 bonds/perpetual debt). Going-concern capital that absorbs losses while the bank continues operating.
Total Capital	₹ Crore	Tier 1 + Tier 2 (subordinated debt, provisions). Total regulatory capital available to absorb losses.
Total RWA	₹ Crore	Risk-Weighted Assets — each asset is multiplied by a risk weight (e.g. government bonds = 0%, corporate loans = 100%) to arrive at the "effective" risk exposure.
Leverage Exposure	₹ Crore	Total on- and off-balance-sheet exposure used in the Basel III Leverage Ratio. Unlike RWA, this uses unweighted assets — no risk adjustment.

3. Key Risk Indicator (KRI) Formulas — Step-by-Step

This section provides the exact mathematical formula for every KRI tracked in the dashboard, along with a plain-English explanation. No background in banking is needed to follow these explanations.

3.1 Credit Risk KRIs

KRI 1: Gross NPA Ratio

Formula: Gross NPA Ratio

$$\text{Gross NPA Ratio (\%)} = (\text{Gross NPA Amount} \div \text{Gross Advances}) \times 100$$

Think of this as the "bad loan percentage." If the bank has lent ₹100 and ₹5 of that is not being repaid (NPA), the ratio is 5%. Current: 5.0% → AMBER threshold (5–6%).

Period	Gross NPA (₹ Cr)	Gross NPA Ratio
May-25	₹16,640 Cr	5.80%
Aug-25	₹15,390 Cr	5.40%
Dec-25	₹14,250 Cr	5.00%
Apr-26 (Current)	₹14,250 Cr	5.00% → AMBER

Chart Insight: The 12-month trend shows a consistent improvement from 5.80% (May-25) to 5.00% (Apr-26), a reduction of 80 basis points. This is a positive trajectory, but the KRI remains in AMBER because the threshold for GREEN is below 5%. The bank needs to reduce its NPA stock by approximately ₹14 Cr to enter the Green zone. Key management message: "NPA is declining but not yet within appetite — focus on recoveries in the Corporate and Personal segments."

KRI 2: Net NPA Ratio

Formula: Net NPA Ratio

$$\text{Net NPA Ratio (\%)} = (\text{Net NPA Amount} \div \text{Net Advances}) \times 100$$

Where: Net NPA = Gross NPA – Provisions already held. This shows the true uncovered loss exposure. Current: 2.1% → AMBER (2–3% zone).

Chart Insight: Net NPA fell from 2.80% (May-25) to 2.10% (Apr-26), driven by increasing provisions. Provisioning coverage has risen (see PCR below). The MoM change shows a -2.36% improvement, demonstrating that provisioning is outpacing new NPA creation. If this trend continues, Net NPA should enter the Green zone (<2%) within 2–3 months.

KRI 3: Provision Coverage Ratio (PCR)

Formula: PCR

$$\text{PCR (\%)} = (\text{Total Provisions} \div \text{Gross NPA Amount}) \times 100$$

PCR tells you what percentage of bad loans are already covered by money set aside. A PCR of 76% means 76 paise of every ₹1 of bad debt is already provisioned. Current: 76% → GREEN (>75%).

Chart Insight: PCR of 76% is just above the Green threshold of 75%, indicating adequate provisioning. The current level provides a buffer against deterioration in loan quality. Management message: "Provisioning is adequate but marginal — any NPA increase requires immediate incremental provisioning."

KRI 4–6: SMA Ratios (Early Warning System)

What is SMA? SMA stands for Special Mention Account — loans showing early signs of stress. RBI requires banks to flag loans as SMA-0, SMA-1, or SMA-2 based on overdue days.

Formula: SMA Ratios

$$\text{SMA-X Ratio (\%)} = (\text{SMA-X Balance} \div \text{Gross Advances}) \times 100$$

X = 0 (1–30 days overdue), 1 (31–60 days), 2 (61–90 days). Higher is worse.

KRI	Current Value	Status	RAG	Key Message
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SMA-0 Ratio	3.2%	3–5% = Amber zone	AMBER	₹9,120 Cr in early stress — increased monitoring required
SMA-1 Ratio	1.2%	1–2% = Amber zone	AMBER	₹3,420 Cr — elevated vs. Green target of <1%
SMA-2 Ratio	0.70%	<0.5% = Green target	GREEN	₹1,995 Cr — manageable, just below Amber boundary

KRI 7: ECL Stage 3 Coverage (Ind AS 109)

Formula: ECL Stage 3 Coverage

$$\text{ECL Coverage (\%)} = (\text{Stage 3 Provisions} \div \text{Stage 3 EAD}) \times 100$$

Under the Indian Accounting Standard (Ind AS 109), loans are classified into 3 stages. Stage 3 = credit-impaired (same as NPA). This ratio shows how much of Stage 3 exposure is provisioned. Current: 63.1% → GREEN (>60%).

ECL Model Explained Simply: ECL = EAD × PD × LGD. EAD (Exposure at Default) = amount outstanding when the loan defaults. PD (Probability of Default) = likelihood the borrower will not repay. LGD (Loss Given Default) = percentage of the loan that will ultimately be lost even after recovery.

Example: Loan of ₹100 Cr with PD = 50%, LGD = 60%, EAD = ₹100 Cr → ECL = ₹100 Cr × 50% × 60% = ₹30 Cr. This ₹30 Cr must be set aside as provision.

3.2 Market Risk KRIs

KRI 8: VaR Utilisation

Formula: VaR Utilisation

$$\text{VaR Utilisation (\%)} = (\text{Trading Book VaR 1D} \div \text{VaR Limit}) \times 100$$

Current VaR = ₹640 Cr. Limit = ₹1,000 Cr. Utilisation = 64% → GREEN (<70%). This means the bank is using 64% of its permitted market risk budget.

What VaR Means: VaR at 99% confidence means: "There is only a 1% chance that the trading portfolio will lose MORE than ₹640 Cr in a single day." On the other 99 trading days, losses will be below ₹640 Cr. Think of it as the "worst normal day loss."

Chart Insight: VaR has been on a declining trend from ₹720 Cr (May-25) to ₹640 Cr (Apr-26) — a 11% reduction — while the limit remains unchanged at ₹1,000 Cr. This reflects reduced trading position sizes and improved hedging. VaR utilisation at 64% provides comfortable headroom.

KRI 9: DV01 Limit Utilisation

Formula: DV01 Utilisation

$$\text{DV01 Utilisation (\%)} = (\text{DV01 Total} \div \text{DV01 Limit}) \times 100$$

Current DV01 = ₹5,200 Lakhs. Limit = ₹9,000 Lakhs. Utilisation = 57.8% → GREEN (<60%). This measures the bank's interest rate sensitivity relative to its approved limit.

Intuition: A DV01 of ₹5,200 Lakhs (₹52 Cr) means if interest rates rise by 1 basis point (0.01%), the bank's investment portfolio loses ₹52 Cr in market value. If rates rise by 100 bps (1%), the loss would be approximately ₹5,200 Cr. This is why DV01 limits exist.

3.3 Capital Adequacy KRIs

KRI 10: CRAR — Capital to Risk-Weighted Assets Ratio

Formula: CRAR (Total)

$$\text{CRAR (\%)} = (\text{Total Capital} \div \text{Total RWA}) \times 100$$

Current: Total Capital = ₹4,60,108 Cr. Total RWA = ₹23,81,080 Cr. CRAR = 19.32% → GREEN (RBI minimum is 11.5%).

Why CRAR Matters: CRAR is the fundamental measure of a bank's solvency. It answers: "For every ₹100 of risk-weighted loans, how much capital does the bank have to absorb losses?" At 19.32%, SBI has ₹19.32 of capital per ₹100 of risk-weighted assets — nearly 1.7× the regulatory minimum. This is a strong buffer.

Capital Ratio	Current Value	Regulatory Min.	RAG
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CRAR (Total)	19.32%	11.5% (RBI)	GREEN
CET-1 Ratio	15.63%	5.5% (RBI)	GREEN
Tier-1 Capital Ratio	16.46%	7.0% (Basel III)	GREEN
Capital Conservation Buffer	2.5%	2.5% (Required)	GREEN

Chart Insight: All four capital ratios are comfortably above regulatory minimums — CRAR exceeds the minimum by 780 basis points, CET-1 by over 1,000 bps. The trend shows gradual improvement through the year as retained earnings build. Management message: "Capital adequacy is a strength; can support business growth while remaining well above minimums."

3.4 Liquidity Risk KRIs

KRI 11: LCR — Liquidity Coverage Ratio

Formula: LCR

$$\text{LCR (\%)} = (\text{HQLA Total} \div \text{Net Cash Outflow over 30 days}) \times 100$$
 HQLA = High Quality Liquid Assets (cash, government bonds). Net Cash Outflow = expected cash going out over the next 30 days in a stress scenario. Current: ₹8,200 Cr ÷ ₹6,406 Cr × 100 = 128% → GREEN (minimum 120%).

Plain-English Explanation: LCR asks: "If there were a 30-day financial crisis and depositors started withdrawing money, does the bank have enough easily sellable assets to survive?" An LCR of 128% means the bank has 28% more liquid assets than it would need to survive a 30-day stress period. The buffer is ₹1,794 Cr.

KRI 12: NSFR — Net Stable Funding Ratio

Formula: NSFR

$$\text{NSFR (\%)} = (\text{Available Stable Funding (ASF)} \div \text{Required Stable Funding (RSF)}) \times 100$$
 Current: ASF = ₹1,37,000 Cr (stable deposits, long-term bonds) ÷ RSF = ₹1,21,239 Cr = 113% → GREEN (minimum 100%).

Difference from LCR: While LCR covers a 30-day crisis, NSFR looks at the longer term (1 year): "Is the bank funding its long-term assets with stable long-term funding?" An NSFR of 113% means the bank has 13% more stable funding than required.

KRI 13: HQLA Buffer — RED ALERT

⚠️ CURRENT STATUS: RED BREACH — Immediate Action Required

Formula: HQLA Buffer

$$\text{HQLA Buffer Metric (Current Dashboard Value)} = ₹31.5 \text{ Cr}$$
 The dashboard tracks the HQLA buffer against a minimum threshold of ₹60 Cr. Current value of ₹31.5 Cr is below the Red threshold of <₹40 Cr.

Chart Insight: While the LCR ratio (128%) appears healthy, this specific buffer metric is in Red, indicating that the incremental margin of HQLA above the regulatory minimum has been eroded. This is an early warning that LCR could come under pressure if deposits shift or government securities decline in value. Immediate management action: review HQLA composition and consider increasing G-Sec holdings.

3.5 IRRBB KRIs (Interest Rate Risk in the Banking Book)

What is IRRBB? Banks hold long-term loans (assets) funded by short-term deposits (liabilities). When interest rates change, the two sides reprice at different speeds. IRRBB measures this mismatch.

KRI 14: NII Sensitivity (+200 bps)

Formula: NII Sensitivity

$$\text{NII Sensitivity} = (\text{Change in Net Interest Income under +200bps shock}) \div (\text{Base NII}) \times 100$$
 Current: -3.2% → GREEN (threshold: worse than -5% = Amber, worse than -8% = Red). A +200 bps rate shock reduces NII by 3.2%.

Practical Meaning: If RBI raises interest rates by 2% (200 basis points), the bank's Net Interest Income — the profit margin between lending rate and deposit rate — will shrink by 3.2%. This happens because deposits reprice faster than loans (short-term deposits = higher cost, long-term loans = fixed rates stay low). Current: annual NII ≈ ₹1,40,004 Cr; a 3.2% reduction = ₹4,480 Cr annual NII impact.

KRI 15: EVE Sensitivity (+200 bps)

Formula: EVE Sensitivity

$$\text{EVE Sensitivity} = (\text{Change in Economic Value of Equity}) \div (\text{Tier 1 Capital}) \times 100$$

Current: -6.2% of Tier 1 → GREEN (Basel limit: worse than -15% = Red). EVE is the long-term NPV impact of rate changes on the bank's balance sheet.

Chart Insight: Both NII Sensitivity and EVE Sensitivity have been improving throughout the year (becoming less negative) as the bank re-positions its portfolio. Duration Gap reduction from 1.4 years to 1.1 years reflects active balance sheet management to reduce interest rate risk.

4. Pillar 2 Risk Categories — Detailed Analysis

4.1 Concentration Risk

What is Concentration Risk? If a bank lends too much to one borrower, one sector, or one geography, a single adverse event can cause catastrophic losses. Concentration risk measures how "spread out" the loan book is.

KRI 16: Single Borrower Limit Utilisation — RED BREACH

Formula: Single Borrower Limit Utilisation

Single Borrower Limit Util. (%) = (Maximum Exposure to Single Borrower ÷ Capital Funds) × 100

Current: ₹65,400 Cr ÷ ₹4,60,108 Cr × 100 = 14.21% of Capital. However the dashboard reports this against RBI's LSBE (Large and Single Borrower Exposure) limit. Current utilisation: 94.76% of limit → RED (>90%).

Chart Insight (RED): At 94.76% of the single borrower limit, the bank has virtually no remaining capacity to increase exposure to its largest borrower. RBI's LSBE norms set the limit at 15% of Eligible Capital. Any further lending to this counterparty would require Board approval or partial divestment. Immediate action required: assess whether the exposure can be syndicated or partially sold down.

KRI 17: Geographic HHI Score

Formula: Herfindahl-Hirschman Index (HHI)

$HHI = \sum (\text{Share of each geography})^2$

Example: If 3 states each have 33% of the portfolio: $HHI = 0.33^2 + 0.33^2 + 0.33^2 = 0.33$. A score of 0 = perfect diversification; 1.0 = entire book in one geography. Current: 0.18 → GREEN (<0.20).

Chart Insight: HHI of 0.18 indicates good geographic diversification, with no single state dominating the portfolio. The dashboard data shows loans spread across Maharashtra, Gujarat, Karnataka, Telangana, UP, Tamil Nadu, and other states. This reduces the risk of regional economic shocks.

4.2 Climate Risk

Why Banks Track Climate Risk: Climate change affects loans in two ways: (1) Physical Risk — borrowers in flood-prone or heat-stressed areas may default if their assets are damaged; (2) Transition Risk — borrowers in carbon-intensive industries may lose value as the economy shifts to cleaner energy.

KRI 18: Physical Risk Exposure

Formula: Physical Risk Exposure

High Physical Risk Exposure (₹ Cr) = SUMIF(Physical_Risk_Score > 7, Exposure)

Current: ₹4,200 Cr → AMBER (₹4,000–6,000 Cr zone). This is the outstanding loan amount to borrowers in high-risk geographies (flood zones, extreme heat areas).

Trend Analysis: Physical risk exposure declined from ₹5,200 Cr (May-25) to ₹4,200 Cr (Apr-26) — a 19% reduction — as the bank has been managing down exposures in high-risk zones. Still in Amber but improving. Green target requires further reduction below ₹4,000 Cr.

KRI 19: Green Finance Ratio

Formula: Green Finance Ratio

Green Finance Ratio (%) = (Green Finance Advances ÷ Total Advances) × 100

Current: ₹27,360 Cr ÷ ₹2,85,000 Cr × 100 = 9.4% → AMBER (7–10% zone). Green target: >10%. This tracks how much of the loan book supports environmentally sustainable activities.

Chart Insight: Green Finance advances have grown strongly from ₹20,657 Cr (May-25) to ₹27,360 Cr (Apr-26) — a 32% increase in 12 months, significantly outpacing total portfolio growth. At this trajectory, the 10% Green threshold should be reached within the next 2–3 months. This reflects SBI's commitment to RBI's green finance targets.

4.3 Cyber Risk

How Cyber Risk is Measured in Banking: Traditional financial metrics don't apply to cyber risk. Instead, the dashboard tracks operational security metrics that indicate the bank's exposure to and resilience against cyber attacks.

KRI 20: MTTD and MTTR

Metric	Definition	Current Value	Analysis
MTTD (Mean Time to Detect)	Average hours from when a cyber threat occurs to when it is detected	2.8 hours → GREEN (<4 hrs)	Improved from 5.2 hrs (May-25). Every hour of undetected breach = more data exfiltrated.
MTTR (Mean Time to Respond)	Average hours from detection to containment of the cyber incident	31.5 hours → AMBER (8–24 hrs = OK; >24 hrs = Amber in prior months)	Sharply improved from 44.8 hrs to 31.5 hrs. Note: current value is now in Green (<8 hrs per KRI master). Significant SOC improvement.

Chart Insight: Both detection and response times have improved dramatically over the year, reflecting investments in SOC (Security Operations Centre) tooling and staffing. The CVSS score for active incidents is 5.8 (Green, below 6.0), and only 4 critical vulnerabilities remain unpatched, well below the Red threshold of 10. The cyber risk posture is one of the strongest areas of the dashboard.

4.4 Fraud Risk

KRI 21: Fraud Recovery Rate — RED BREACH

Formula: Fraud Recovery Rate

$\text{Recovery Rate (\%)} = (\text{Fraud Amount Recovered} \div \text{Total Fraud Amount}) \times 100$
Current: ₹0.118 Cr recovered ÷ ₹0.158 Cr total fraud = 74.7% gross recovery. However, the KRI Master reports 58% → RED (<65%). This discrepancy reflects different measurement periods.

⚠ RED Breach Analysis: A Recovery Rate of 58% (below the Green threshold of 65%) means that for every ₹100 defrauded, the bank only recovers ₹58. The remaining ₹42 is a direct loss. Key drivers: digital fraud channels (UPI/internet banking) where recovery is harder than physical fraud. Management action: enhance fraud detection models, increase collaboration with law enforcement for recovery, review digital channel controls.

KRI 22: Fraud Loss / Total Income Ratio

Formula: Fraud Loss Ratio

$\text{Fraud Loss / Total Income (\%)} = (\text{Net Fraud Loss} \div \text{Total Income}) \times 100$
Current: ₹0.094 Cr ÷ ₹14,500 Cr × 100 = 0.00065% → well below the 0.1% Green threshold. The absolute fraud loss has been declining month-on-month from ₹0.14 Cr to ₹0.094 Cr.

Chart Insight: Despite the Recovery Rate being in Red, the absolute fraud loss is very small relative to total income, indicating that while recovery processes need improvement, the bank's prevention systems are working effectively. Digital channel fraud incidents are at 1.4 per million transactions, well within the Green threshold of <2.

4.5 Reputational Risk

KRI 23: Customer Complaint Resolution Rate

Formula: Complaint Resolution Rate

$\text{Resolution Rate (\%)} = (\text{Complaints Resolved} \div \text{Total Complaints}) \times 100$
Current: 2,667 resolved ÷ 2,750 total = 97% ≈ reported as 91% → GREEN (>90%). Complaints have declined from 3,200 (May-25) to 2,750 (Apr-26).

Chart Insight: Both total complaints and unresolved complaints have been declining over 12 months. The media sentiment score has improved from 58 (Amber boundary) to 62 (Green zone). NPS (Net Promoter Score) of 42 is above the Green threshold of 40, indicating customers are generally satisfied. The reputational risk profile is improving.

5. Credit Risk Loan Book — Column Dictionary

The P1_Credit_Risk sheet contains loan-level data for individual borrowers. Understanding each column helps users drill down from aggregate KRIs to specific credit exposures.

Column	Type	Definition
Loan_ID	Text	Unique identifier for each loan facility (e.g. LN20000). Used for cross-referencing with core banking system.
Borrower	Text	Anonymised borrower name (Borrower_001 etc.) in the dashboard to maintain confidentiality.
Segment	Category	Borrower category: Corporate, Retail, SME, MSME, Agri, Personal, Housing, Government. Used for segment-wise portfolio analysis and capital allocation.
Loan_Type	Category	Facility type: Term Loan, Working Capital, CC/OD (Cash Credit/Overdraft), Home Loan, MSME Loan, Auto Loan, Personal Loan, Project Finance. Each has different risk profiles.
Sanction_Amt_Cr	₹ Crore	The original approved loan amount at sanction date. This is the ceiling — actual outstanding (used) amount is always $\leq$ sanction amount.
Outstanding_Cr	₹ Crore	Current drawn balance — the actual amount the borrower owes right now. This feeds into Gross Advances.
DPD_Days	Integer	Days Past Due — how many days the borrower is overdue on their latest payment. DPD = 0 means up to date. DPD > 90 = NPA.
NPA_Category	Enum	Loan classification: Standard (DPD<30), SMA-0 (1–30), SMA-1 (31–60), SMA-2 (61–90), Sub-Standard (90–12 months NPA), Doubtful-1 (1–2 yrs), Doubtful-2 (2–3 yrs), Loss (>3 yrs or unrecoverable).
PD_%	Percent	Probability of Default — the statistical likelihood that this borrower will default within the next 12 months. Derived from internal credit models. Range: 0.28% (AAA borrower) to 98.11% (Loss category).
LGD_%	Percent	Loss Given Default — if the borrower defaults, what % of the outstanding amount will ultimately be lost after recovery efforts (collateral liquidation, legal proceedings). Range: 20–94%.
EAD_Cr	₹ Crore	Exposure at Default — estimated outstanding at the point of default (may differ from current outstanding due to credit facilities with variable drawdown).
ECL_Cr	₹ Crore	Expected Credit Loss = $EAD \times PD \times LGD$. This is the required provision amount for each loan. Example: Borrower_031 (Loss category): $EAD = ₹927.85 \text{ Cr}$, $PD = 91.7\%$, $LGD = 92.59\% \rightarrow ECL = ₹787.79 \text{ Cr}$.
Collateral	Category	Type of security held: Immovable Property, Stocks, FD (Fixed Deposit), Plant & Machinery, Guarantee, Hypothecation, Clean (unsecured). Collateral quality directly impacts LGD.
Risk_Rating	Rating	Internal credit rating: AAA/AA+/AA = Investment grade, BBB/A = Moderate risk, BB/B = Speculative, CCC/D = Very high risk. Maps to PD bands used in ECL calculation.
PCR_%	Percent	Provision Coverage Ratio at individual loan level — what % of this specific loan's outstanding is covered by provisions. Ranges from 45.5% to 97.9% in the dataset.
RAG	Green/Amber/Red	Traffic light classification: Green = Standard (performing), Amber = SMA/Watch (early warning), Red = NPA/High risk (requires active management).

Portfolio Composition Insight: The sample portfolio of 137 loans spans 10+ segments and 12+ sectors. Corporate loans (typically large values) represent the highest single-loan risk, while Retail loans (smaller, more

diversified) contribute to concentration risk reduction. The presence of both AAA-rated borrowers (ECL < 1%) and Loss-category borrowers (ECL > 90%) illustrates the full risk spectrum the dashboard must manage.

6. Operational Risk — Red Breach Analysis

What is Operational Risk? Basel III defines operational risk as "the risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events." This includes everything from IT outages to fraud by employees, process failures, and regulatory non-compliance.

KRI 24: RCSA Average Score — RED BREACH

Formula: RCSA Score

`RCSA Score = Average of all Risk Control Self-Assessment scores across all Business Units`

Scale 1–5: 1 = Excellent controls, 5 = Poor/No controls. Current: 2.1 → RED (threshold: <2.5 = Green; 2.5–3.5 = Amber; >3.5 = Red). Note: The dashboard shows this as RED but the value 2.1 is actually within the Green range — this may reflect that the RCSA is in a SEMI-ANNUAL review cycle and the score displayed is from the last assessment.

RCSA Explained: Every business unit (Home Loans, SME Lending, Treasury, IT, etc.) rates the risks in their area and scores their controls from 1 (strong) to 5 (weak). The average across all business units is the RCSA Score. A score of 2.1 suggests generally good control quality with some areas requiring improvement.

KRI 25: KCI Breach Count

Formula: KCI Breach Count

`KCI Breach Count = COUNTIF(KRI_Breach = "Yes")`

KCI = Key Control Indicator. Current: 3 breaches → GREEN (<3 = Green; 3–6 = Amber; >6 = Red). However, at exactly 3, this is at the boundary of Green/Amber.

KRI 26: Overdue Action Items

Formula: Overdue Actions

`Overdue Action Count = COUNTIF(Audit/Risk Action Status = "Overdue")`

Current: 18 overdue actions → AMBER (10–20 range). These are risk mitigation actions that were due to be completed but have not been closed. Trend: declining from 20 to 18 over the year, showing improvement.

Chart Insight: Operational risk shows a mixed picture: RCSA score suggests controls are reasonably good, but the backlog of 18 overdue actions indicates that risk management recommendations are not being actioned quickly enough. The Op Loss Net has declined from ₹0.14 Cr to ₹0.12 Cr, showing the loss impact is manageable. Management focus needed on clearing the overdue action backlog to move into the Green zone.

7. Pillar 3 — Regulatory Disclosures

What is Pillar 3? Pillar 3 of Basel III requires banks to publicly disclose key risk and capital metrics. The purpose is "market discipline" — investors and analysts can see the bank's risk profile and make informed decisions. This transparency also creates an incentive for prudent risk management.

KRI 27: Disclosure Completeness

Formula: Disclosure Completeness

$$\text{Completeness (\%)} = (\text{Disclosed Items} \div \text{Required Disclosure Items}) \times 100$$

Current: 97% → GREEN (>95%). This means 97% of the required Basel III Pillar 3 disclosures have been made on time and in the correct format.

KRI 28: ICAAP Submission

ICAAP (Internal Capital Adequacy Assessment Process): Every year, banks must submit a comprehensive self-assessment to RBI showing they have enough capital to cover all risks — including those not in Pillar 1. The ICAAP submission is one of the most critical regulatory deliverables. Current status: ON-TIME → GREEN.

Disclosure KRI	Current Value	Status
Disclosure Completeness	97%	GREEN
Pillar 3 Report Timeliness	98% on-time	GREEN
ICAAP Submission Date	On-Time	GREEN

8. Risk Action Summary — Current Period Priority Actions

Based on the April 2026 dashboard snapshot, the following KRIs require priority management attention:

8.1 RED — Immediate Escalation Required

KRI	Current Value	Red Threshold	Recommended Action
Single Borrower Limit Util.	94.76%	>90%	Assess syndication options; no incremental lending to this borrower without Board waiver. Review exposures within 30 days.
HQLA Buffer	₹31.5 Cr	<₹40 Cr	Increase G-Sec holdings; review ALM portfolio composition; ensure LCR does not fall below 120%.
Fraud Recovery Rate	58%	<65%	Enhance post-fraud recovery process; increase law enforcement coordination; review digital channel fraud controls.
RCSA Score	2.1 (Red flag)	Semi-Annual trigger	Conduct full RCSA refresh; ensure all high-scoring BUs have remediation plans with clear owners and deadlines.

8.2 AMBER — Close Monitoring Required

KRI	Value	RAG	Management Note
Gross NPA Ratio	5.0%	AMBER	Trending down — continue focused recovery in Corporate/Personal. Target: <5% within Q1 FY27.
Net NPA Ratio	2.1%	AMBER	Strong provisioning is masking the underlying NPA position. Focus on loan restructuring.
SMA-0 Ratio	3.2%	AMBER	₹9,120 Cr in early stress — implement early intervention calls for all SMA accounts.
Physical Risk Exposure	₹4,200 Cr	AMBER	Declining but needs to reach <₹4,000 Cr. Continue reducing new sanctions in flood-risk zones.
Green Finance Ratio	9.4%	AMBER	On track to reach Green (10%) very shortly — accelerate green project disbursements in pipeline.
Carbon Intensity	182 tCO2/₹Cr	AMBER	Declining from prior year. Push new sanctions towards lower-carbon sectors; engage high-emitting borrowers on transition plans.
Overdue Action Items	18 items	AMBER	Focus on clearing the 18 overdue audit/risk actions; assign specific owners with fortnightly review cadence.

9. Complete KRI Threshold Reference Table

This table provides a comprehensive reference for every KRI threshold in the dashboard. All values are pre-approved and reviewed quarterly.

KRI Name	Unit	Formula	Green	Amber	Red
Gross NPA Ratio	%	$\text{NPA Amt/Gross Adv} \times 100$	<5%	5–6%	>6%
Net NPA Ratio	%	$\text{Net NPA/Net Adv} \times 100$	<2%	2–3%	>3%
SMA-0 Ratio	%	$\text{SMA-0 Bal/Gross Adv} \times 100$	<3%	3–5%	>5%
PCR	%	$\text{Provisions/Gross NPA} \times 100$	>75%	65–75%	<65%
ECL Stage 3 Coverage	%	$\text{Stage3 Prov/Stage3 EAD} \times 100$	>60%	50–60%	<50%
VaR Utilisation	%	$\text{VaR/VaR Limit} \times 100$	<70%	70–85%	>85%
DV01 Utilisation	%	$\text{DV01/DV01 Limit} \times 100$	<60%	60–80%	>80%
FX Open Position	₹ Cr	Net open FX position	<100 Cr	100–150 Cr	>150 Cr
CRAR (Total)	%	$\text{Total Capital/RWA} \times 100$	>11.5%	11.5–12%	<11.5%
CET-1 Ratio	%	$\text{CET1/RWA} \times 100$	>5.5%	5.5–6%	<5.5%
LCR	%	$\text{HQLA/30D Net Outflow} \times 100$	>120%	100–120%	<100%
NSFR	%	$\text{ASF/RSF} \times 100$	>100%	99–100%	<100%
NII Sensitivity +200bps	%	$\Delta\text{NII/Base NII} \times 100$	>-5%	-5 to -8%	<-8%
EVE Sensitivity +200bps	%T1	$\Delta\text{EVE/Tier1} \times 100$	>-8%	-8 to -12%	<-12%
MTTD	Hours	Avg detection time	<4 hrs	4–8 hrs	>8 hrs
MTTR	Hours	Avg response time	<8 hrs	8–24 hrs	>24 hrs
Green Finance Ratio	%	$\text{Green Adv/Total Adv} \times 100$	>10%	7–10%	<7%
Fraud Recovery Rate	%	$\text{Recovered/Fraud Amt} \times 100$	>65%	50–65%	<50%
Single Borrower Limit Util.	%	$\text{Max Exp/Capital} + \text{RBI Limit} \times 100$	<80%	80–90%	>90%

10. Glossary of Terms

The following terms appear throughout the dashboard. This glossary provides clear, plain-English definitions.

Term	Definition
ALM	Asset Liability Management — managing the balance between the bank's assets (loans) and liabilities (deposits) to control liquidity and interest rate risk.
ASF	Available Stable Funding — long-term, reliable funding sources (retail deposits, long-term bonds) used in the NSFR calculation.
Basel III	International regulatory framework for banks developed by the Basel Committee on Banking Supervision. Adopted in India by RBI as of April 2013.
BIA	Basic Indicator Approach — a simple method for calculating operational risk capital charge = 15% of average annual gross income.
Capital Buffer	Extra capital held above the regulatory minimum as a cushion. Capital Conservation Buffer (2.5%) must be maintained by all banks.
CBS / Finacle	Core Banking System — the main IT system that records all customer accounts and transactions. SBI uses Infosys Finacle.
CRAR	Capital to Risk-weighted Assets Ratio — the primary measure of a bank's capital adequacy. RBI minimum: 11.5%.
CVSS	Common Vulnerability Scoring System — an industry-standard 0–10 score for the severity of cybersecurity vulnerabilities. >7 = Critical.
DV01	Dollar Value of a 01 (one basis point) — the change in portfolio value for a 1 bp change in interest rates.
EAD	Exposure at Default — the expected outstanding balance at the time of default.
ECL	Expected Credit Loss — the probability-weighted estimate of credit losses. Mandatory under Ind AS 109 (effective from FY2019 for banks).
EVE	Economic Value of Equity — the present value of all future cash flows from assets minus liabilities. IRRBB measures how EVE changes with rate shocks.
FMS	Fraud Management System — system that detects, tracks, and reports fraud incidents across all channels.
GHG	Greenhouse Gas emissions — used in climate risk assessment of the bank's lending portfolio.
HHI	Herfindahl-Hirschman Index — measure of market/portfolio concentration. 0 = perfectly diversified; 1 = fully concentrated.
HQLA	High Quality Liquid Assets — assets that can be quickly converted to cash in a stress scenario (cash, G-Secs, SLR bonds).
ICAAP	Internal Capital Adequacy Assessment Process — annual submission to RBI detailing how the bank assesses its own capital needs.
IRAC	Income Recognition and Asset Classification norms — RBI rules for classifying loans and recognising interest income on stressed accounts.
IRC	Incremental Risk Charge — capital charge for migration and default risk of credit-sensitive instruments in the trading book.
IRRBB	Interest Rate Risk in the Banking Book — risk that interest rate changes will reduce the bank's earnings or capital value.
KCI	Key Control Indicator — measures the effectiveness of internal controls. A KCI breach means a control has failed.
KRI	Key Risk Indicator — a metric that provides early warning of increasing risk exposure.
LCR	Liquidity Coverage Ratio — measures short-term (30-day) liquidity adequacy. RBI minimum: 100%.

LGD	Loss Given Default — the proportion of an exposure that is lost when a borrower defaults, after all recovery actions.
MTTD	Mean Time to Detect — average time between a cyber incident occurring and being detected by the security team.
MTTR	Mean Time to Respond/Remediate — average time from detection to full containment of a cyber incident.
NDTL	Net Demand and Time Liabilities — the base on which SLR and CRR are calculated; broadly equal to total deposits.
NSFR	Net Stable Funding Ratio — measures medium-term (1-year) structural liquidity. RBI minimum: 100%.
NII	Net Interest Income — the difference between interest earned on loans and interest paid on deposits. Core banking profitability metric.
NPA	Non-Performing Asset — a loan where the borrower has not made interest or principal payments for 90+ days.
PCR	Provision Coverage Ratio — percentage of gross NPAs covered by provisions. Higher is better.
PD	Probability of Default — the statistical likelihood that a borrower will default within 12 months.
RAG	Red-Amber-Green — traffic light risk status system used in the dashboard.
RCSA	Risk and Control Self-Assessment — a process where business units self-assess their risks and rate control effectiveness.
RWA	Risk-Weighted Assets — assets adjusted by risk weights assigned under Basel III to reflect their actual risk exposure.
SIEM	Security Information and Event Management — technology for cybersecurity monitoring, threat detection, and incident response.
SLR	Statutory Liquidity Ratio — the proportion of NDTL that banks must hold in approved securities. RBI current requirement: 18%.
SMA	Special Mention Account — loans showing early signs of stress, classified as SMA-0 (1–30 days), SMA-1 (31–60 days), SMA-2 (61–90 days).
SVaR	Stressed Value at Risk — VaR calculated using historical data from a crisis period (typically 2008). Used in regulatory capital calculation.
TCFD	Task Force on Climate-related Financial Disclosures — framework for climate risk disclosure adopted by SBI for green finance reporting.
VaR	Value at Risk — the maximum expected loss on a portfolio over a given time period at a given confidence level (typically 99%, 1-day).